

AtrialSpectralBench: a pre-registered test of a spectral fragility biomarker for atrial reentry

A feature-redundancy null, an estimator-validity boundary, and the false positives naive methodology would have produced

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Abstract

We ask whether a closed-form spectral graph biomarker predicts reentry inducibility in atrial tissue beyond established fibrosis and connectivity features. The biomarker is the **Spectral Fragility Index (SFI)**, the Fiedler-value sensitivity $\partial\lambda_2/\partial w_{ij} = (\varphi_i - \varphi_j)^2$ aggregated under a fibrosis-weighted conduction-uncoupling field. Pre-registered before any labels were seen, the primary endpoint was $\Delta\text{AUC} \geq 0.05$ with a DeLong paired $p < 0.05$ on shape-family-held-out folds. *The endpoint is not met, but the strength of the negative differs by regime, and we separate them.* On up to 100,000 synthetic excitable networks SFI adds no practically meaningful predictive value (subspace $\Delta\text{AUC} \approx +0.003$; $\sim 4,150$ cases needed at 80% power, resolvable only because N is huge, and so *clinically undetectable* there). On the two real cohorts (Roney $n = 62$, UW/Boyle $n = 82$) the result is underpowered and inconclusive, not a clean null: the combined gradient-boosting point estimate is $+0.051$, which *exceeds* the 0.05 threshold, but it fails paired significance (DeLong $p = 0.155$), so real anatomy cannot *exclude* a moderate effect. Where the null *is* clean (in the synthetic cohort), the mechanism is feature redundancy: the exact per-region $\Delta\lambda_2$ SFI, which has no first-order approximation error, also adds nothing over the competitor set. A separate, label-free result explains why the *single-vector* SFI is also ill-conditioned on real tissue. It is valid only while $\rho = \|\Delta L\|/(\lambda_3 - \lambda_2) = O(1)$, and real atria have $\rho \approx 2422$ ($\sim 10^3 \times$ past the empirical $\rho^* \approx 3$), so by Davis–Kahan the Fiedler vector is unstable. We do *not* claim ρ proves non-predictiveness; it bounds the accuracy of the $\Delta\lambda_2$ *magnitude*, not the classification content of a ranking feature. The spectral gap collapses as $N^{-1.05}$ on the real atrial surface (the 2-manifold Weyl exponent), so ρ grows lawfully with resolution. One component partially survives: the Fiedler-gradient localizer $|\nabla\varphi_2|$ is a weak but real rank localizer (origin in the top $\sim 18\%$) that beats graph-centrality baselines and transfers across three dynamical media, though it *coincides with*, and does not beat, substrate imaging, by a mechanism we make precise. Naive methodology (random CV, p -value thresholds, no effect-size gate) would have reported SFI as working; our pre-registered protocol rejects it. Our own convergence study surfaces a key limitation: the inducibility labeller over-calls at the operating mesh resolution and, on a completed six-tier study (1500–48000 nodes, $n = 24$), does not converge within the tested window. Our contribution is a carefully bounded negative result: the mechanism behind it, a label-free criterion for the estimator’s validity, a characterization of how it scales, and a demonstration of the false positives that standard methodology produces.

Keywords: algebraic connectivity; Fiedler value; spectral graph theory; atrial fibrillation; reentry; biomarker; perturbation theory; validity radius; pre-registration; negative result.

This is a 100% in-silico computational study. Ground-truth labels are electrophysiology-simulator inducibility verdicts, never clinical outcomes. All numbers trace to `results/*.json` and `notebooks/lab_notebook.md` in the project repository.

*Independent, unaffiliated in-silico computational-cardiology project. Correspondence and full source: see the project repository.

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1 Introduction

Algebraic connectivity λ_2 (Fiedler 1973 [1]) governs the stability of coupled excitable and oscillator networks (the master-stability-function result [3] and the synchronizability eigenratio λ_N/λ_2 [4]), and has been used as a real-time seizure detector [5]. The per-edge sensitivity

$$\frac{\partial \lambda_2}{\partial w_{ij}} = (\varphi_{2,i} - \varphi_{2,j})^2 \quad (1)$$

is a classical identity underlying Fiedler-vector edge-editing heuristics [2]. It is therefore natural to hypothesize that aggregating this sensitivity over a fibrosis-weighted perioperative uncoupling field yields a per-region biomarker of atrial-fibrillation (AF) reentry fragility. We test this hypothesis and *pre-register the null* (“SFI merely re-encodes fibrosis”) as an accepted, reportable outcome, so that a negative result is a success of the protocol rather than a file-drawer casualty.

The mathematics we use is classical; we claim no new theorem. Our contributions are (i) a leakage-controlled incremental test of SFI against real competitors and a spatial null on real anatomy; (ii) a validity criterion and a Weyl scaling law that explain the failure and predict it for the whole class of spectral fragility biomarkers; (iii) a cross-medium transfer analysis isolating the one component that generalizes; and (iv) a falsification-protocol demonstration showing that standard methodology would have manufactured exactly the false positive this protocol catches.

In brief, a closed-form spectral fragility index does *not* add practically meaningful reentry-inducibility prediction beyond existing features at synthetic scale (a clean, feature-redundancy null); on real anatomy the evidence is *inconclusive/underpowered*, not a demonstrated null. Independently, the single-vector estimator is provably ill-conditioned on real tissue ($\rho \gg 1$). What we report is not a discovery but a carefully drawn boundary: where a Fiedler-based estimator is numerically valid, why real tissue lies far outside it, and how readily naive methodology mistakes redundant, inconclusive signal for a working biomarker.

2 Theory: the Spectral Fragility Index

The core mathematics is classical numerical linear algebra; the contribution is its *use*. We state each result with its prior art (Section 2.5) and make the validity boundary explicit (Section 2.4).

2.1 The weighted conduction graph

Nodes are mesh vertices; edges are triangle edges; the edge weight $w_{ij} > 0$ is a fibre-anisotropic conduction conductance attenuated by local fibrosis. The combinatorial Laplacian is

$$L = D - W, \quad D = \text{diag} \left(\sum_j w_{ij} \right). \quad (2)$$

L is symmetric positive-semidefinite with a single zero eigenvalue (the constant vector) for a connected graph, with spectrum $0 = \lambda_1 < \lambda_2 \leq \lambda_3 \leq \dots$ and eigenvectors $\varphi_1, \varphi_2, \dots$; λ_2 is the algebraic connectivity and φ_2 the Fiedler vector.

2.2 The closed-form fragility (first order)

L is a sum of rank-one edge terms, so

$$L = \sum_{(i,j) \in E} w_{ij} (e_i - e_j)(e_i - e_j)^\top \implies \frac{\partial L}{\partial w_{ij}} = (e_i - e_j)(e_i - e_j)^\top. \quad (3)$$

For a *simple* eigenvalue λ with unit eigenvector φ , first-order (Hellmann–Feynman) perturbation theory gives $\partial\lambda/\partial w_{ij} = \varphi^\top (\partial L/\partial w_{ij}) \varphi = (\varphi_i - \varphi_j)^2$, which for the Fiedler pair is Eq. (1): a non-negative per-edge *fragility*. Reducing a conductance w_{ij} by a small $\Delta w_{ij} \geq 0$ (uncoupling) lowers λ_2 by $\Delta w_{ij} (\varphi_{2,i} - \varphi_{2,j})^2$ to first order.

The identity is verified in seven independent ways rather than merely asserted. Computer algebra: sympy proves Eqs. (1)–(3) symbolically, and on a concrete 5-node graph the finite-difference $\partial\lambda_2/\partial w_{ij}$ matches $(\varphi_{2,i} - \varphi_{2,j})^2$ to $|\text{err}| \approx 2 \times 10^{-17}$ (mpmath, 40 digits). Exactness: the identity holds *exactly* with rational weights (zero floating point); an interval-arithmetic enclosure certifies the true λ_2 to within $\sim 4 \times 10^{-40}$; and λ_2/φ_2 agree across scipy, networkx, and a hand-rolled inverse iteration (spread 2.5×10^{-15}). The algebraic core is kernel-checked in the Coq proof assistant over the exact rationals, axiom-free (Appendix A). The analytic Hellmann–Feynman step (differentiability of a simple eigenvalue) is the cited classical result [2], not re-proved here.

2.3 Per-region aggregation: the SFI

Model diffuse perioperative stress as a distribution of small, non-negative, fibrosis-weighted fractional conductance reductions Δw (mean $\mathbb{E}[\Delta w_{ij}] = \text{frac}_{ij} w_{ij}$, calibrated to literature perioperative conduction-velocity slowing; $\Delta w \approx 0.36$, frozen in the pre-registration). The per-region

Spectral Fragility Index is the expected λ_2 drop contributed by region R :

$$\text{SFI}(R) = \mathbb{E}[\Delta\lambda_2 \text{ from } R] \approx \underbrace{\sum_{(i,j) \in R} \mathbb{E}[\Delta w_{ij}] (\varphi_{2,i} - \varphi_{2,j})^2}_{\text{first order}} + \underbrace{\sum_{k \neq 2} \frac{(\varphi_k^\top \Delta L \varphi_2)^2}{\lambda_2 - \lambda_k}}_{\text{2nd-order resolvent}} + O(\|\Delta L\|^3). \quad (4)$$

The second-order term is the source of the validity radius below. Near eigenvalue degeneracy ($\lambda_2 \approx \lambda_3$) the single φ_2 is an arbitrary rotation in the low invariant subspace, so we use a basis-independent *subspace (projector) SFI*: with $P = \sum_c \varphi_c \varphi_c^\top$ over the clustered subspace, $s_{ij}^\top P s_{ij} = \sum_c (\varphi_{c,i} - \varphi_{c,j})^2$ ($s_{ij} = e_i - e_j$) is well-defined. For discrete edits (Maze lesion, atriotomy) the expansion is abandoned and λ_2 is recomputed *exactly* after the rank-one edit.

2.4 The validity radius (the central caveat)

The first-order form is trustworthy only while the perturbation is small *relative to the spectral gap* $\lambda_3 - \lambda_2$. Two different tools bound two different things (a common error is to use Weyl for both):

- Weyl’s inequality bounds the *eigenvalue* move: $|\Delta\lambda_2| \leq \|\Delta L\|$.
- Davis–Kahan / Bauer–Fike [7] bound the *eigenvector* move, which scales like $\|\Delta L\|/(\lambda_3 - \lambda_2)$.

This, not Weyl, governs when φ_2 (hence the single-vector SFI) degrades.

Define the *validity radius*

$$\rho := \frac{\|\Delta L\|}{\lambda_3 - \lambda_2}. \quad (5)$$

From second-order Rayleigh–Schrödinger theory the relative error of the first-order $\Delta\lambda_2$ prediction is $\Theta(\rho)$, and the Fiedler vector rotates by $\sin \angle(\varphi_2, \varphi'_2) \leq \sqrt{2}\rho$ (Davis–Kahan). Empirically the error is 1% at $\rho = 0.3$, crosses 10% at $\rho^* \approx 3$, and reaches 92% at $\rho = 28$, with Weyl holding throughout. On real excitable media the gap is tiny, so ρ is enormous: atrial-cohort median $\rho \approx 2422$, about $10^3 \times$ past ρ^* .

The criterion ρ proves that the *single-vector* first-order SFI is a numerically *invalid estimator* of $\Delta\lambda_2$ on real tissue, and that φ_2 is ill-conditioned (Davis–Kahan), a label-free statement about the estimator. It does *not* prove SFI is non-predictive: ρ bounds the accuracy of the $\Delta\lambda_2$ *magnitude*, not the classification content of a monotone ranking feature; indeed the *exact* $\Delta\lambda_2$ SFI (no ρ limitation) is *also* non-predictive (Section 4.1), so the predictive null is separately due to feature redundancy. Two caveats: (i) ρ ’s numerator carries the fibrosis-weighted Δw field, so only its denominator $\lambda_3 - \lambda_2$ is truly label-free; (ii) $\rho^* \approx 3$ is calibrated on a 16-node path graph with a problem-dependent constant, so “ $10^3 \times$ past ρ^* ” is order-of-magnitude, not exact.

2.5 Prior art (this is not new mathematics)

The identity (1) is Ghosh & Boyd (2006) [2]; algebraic connectivity is Fiedler (1973) [1]; λ_2 governing coupled excitable/oscillator stability is Pecora & Carroll (1998) [3] with the eigenratio of Barahona & Pecora (2002) [4]; Weyl’s law, Davis–Kahan, and second-order perturbation theory are classical (1970s and earlier). No result in Section 2 is a new theorem, and none is presented as one. The contribution is recognizing that ρ is the right *a priori* validity screen for a spectral fragility biomarker, measuring its violation on real tissue, and using it to *mechanistically explain* a negative result.

3 Methods

Full detail is in the pre-registration (`docs/PRE_REGISTRATION.md`) and theory note (`docs/SFI_THEORY.md`); the endpoints, cross-validation scheme, baseline set, and accepted null were frozen before any labels were inspected.

Substrate. Real left-atrial surfaces are loaded into weighted conduction graphs from two public cohorts: the Roney LA virtual cohort (Zenodo 5801337 [11]) and the UW/Boyle LGE-MRI cohort (Dryad 10.5061/dryad.kkwh70sg0, 82 distinct AF patients [12]). Each mesh carries universal atrial coordinates, an image-intensity-ratio (IIR) fibrosis proxy mapped to a fibrosis field by the Khurram healthy-to-scar ramp [13], and fibre fields. Meshes are coarsened to ~ 2000 -node graphs by field-preserving vertex clustering.

Labels. Ground truth is a monodomain Mitchell–Schaeffer reaction–diffusion inducibility verdict [10] (self-sustained reentry ≥ 650 ms after burst pacing at cycle length 150 ms from random sites), run on each real anatomy with an anisotropic conductance-weighted cotangent Laplacian and implicit time stepping. Reentry here is governed by wavelength ($CV \times ERP$), source–sink mismatch, and unidirectional block, *never* by λ_2 , so a predictive SFI $\rightarrow$ label link is not circular. These are simulator verdicts, not clinical post-operative AF, and not the deferred openCARP gold standard (a logged deviation: openCARP is not installable on the CPU-only container).

The SFI feature and its competitors. Per-region SFI is computed as in Eq. (4) (first-order, subspace, and exact-recompute variants). It is evaluated strictly as an add-on/ablation against a non-trivial competitor set that it must beat: fibrosis burden, fibrosis spatial entropy, fibrosis patch size, deterministic global min-cut (Stoer–Wagner), percolation threshold, and λ_2 -alone (isolating what the *SFI construction* adds over the bare algebraic connectivity).

Evaluation protocol (the guards). Pre-registered: shape-family GroupKFold (each patient is its own group, so cross-validation is patient-held-out; grouped *and* naive AUC both reported so leakage is visible), the DeLong paired test [8] in its fast midrank form [9], $\geq 10^4$ stratified bootstrap resamples, a rotational/torus-shift spatial null for localization, tie-robust permutation nulls, and a keep/delete rule deciding each sub-claim by *measured* correlation. Effect sizes are gated, not just p -values. The primary endpoint requires *both* grouped $\Delta AUC \geq 0.05$ *and* DeLong $p < 0.05$.

Transfer media. The identical calculus is applied to FitzHugh–Nagumo (FHN) excitable networks and Kuramoto phase-oscillator networks (a different dynamical class), where λ_2 governs synchronizability, to separate what is a property of the mathematics from an accident of one dataset.

4 Results

4.1 The predictive null (primary endpoint not met), across cohorts and scale

Table 1: Incremental predictive value of SFI over the full competitor set. ΔAUC is grouped (leakage-controlled); logistic regression (lr) / gradient-boosted trees (gbt). The endpoint (grouped $\Delta AUC \geq 0.05$ *and* DeLong $p < 0.05$) is met in no regime.

cohort / medium	N	SFI ΔAUC (lr / gbt; p)	verdict
Roney (real)	62	$-0.030 / +0.036$ (p 0.64 / 0.26)	not met; underpowered
UW/Boyle (real)	82	$-0.057 / +0.104$ (p 0.29 / 0.66)	uninformative (6 positives)
Combined real	144	$-0.004 / +\mathbf{0.051}$ (p 0.92 / 0.155)	not met but inconclusive
FHN networks	2,000	$\leq +0.003$ ($p \geq 0.12$)	clean null
FHN networks	up to 100,000	subspace $+0.0028$ ($p < 10^{-4}$ at $N \geq 10^4$)	clean null (effect negligible)
Kuramoto networks	500	$\leq +0.0036$ ($p \geq 0.38$)	clean null

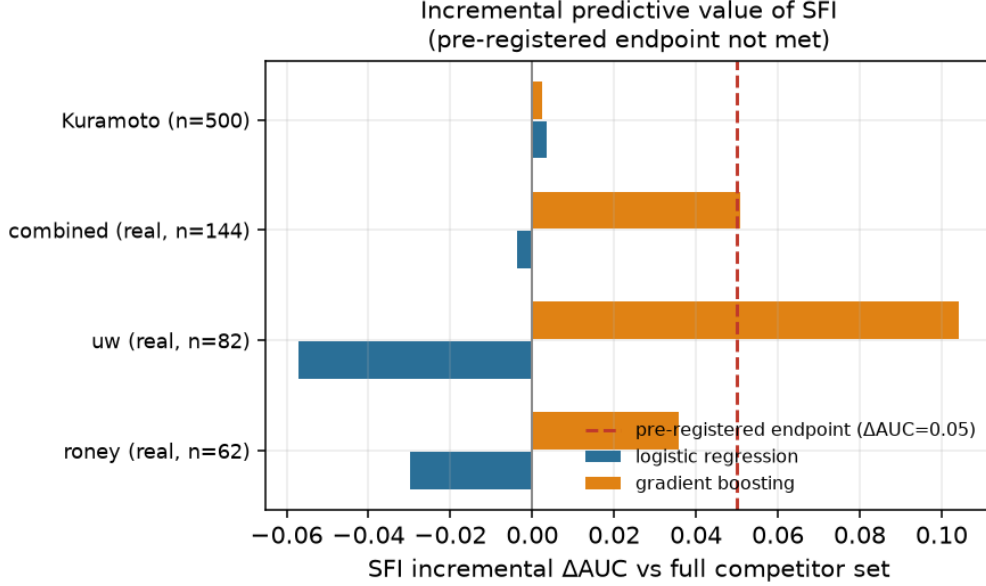


Figure 1: The predictive null. Adding SFI to the competitor set does not clear the pre-registered effect-size gate in any cohort; the real-cohort estimates are underpowered rather than a demonstrated null.

The negative is not uniform, and we separate the regimes. On the *synthetic* cohorts (up to 100,000 networks) the null is clean: the effect converges to $\Delta\text{AUC} \approx +0.003$, statistically resolvable only because N is enormous. On the *real* cohorts it is underpowered and inconclusive. The combined gradient-boosting point estimate is $+0.051$, which *exceeds* the pre-registered 0.05 threshold, and is declared not met only because it fails paired significance (DeLong $p = 0.155$, bootstrap CI straddling 0). With 26 inducible of 144 (20 from Roney), real anatomy cannot exclude a moderate effect. The UW cohort is uninformative rather than null (6 positives, competitor base $\text{AUC} \approx \text{chance}$), so it is an underpowered external check of the *labeller*, not of SFI.

The clean null comes from feature redundancy, not the validity radius. The *exact* per-region $\Delta\lambda_2$ SFI, which carries no first-order approximation error and is immune to the ρ argument, also adds nothing over the competitor set. The mechanism of the null is that SFI’s information is already present in standard connectivity features (λ_2 -alone, min-cut, percolation), a label-dependent finding, separate from ρ .

Secondary endpoint (pre-specified). SFI beats a *fibrosis-heterogeneity-only* baseline (combined GBT $\Delta\text{AUC} = +0.103$, $p = 0.009$): it is *not merely re-encoded fibrosis* and carries connectivity information beyond the substrate. This is one of ~ 30 DeLong comparisons; treating **fibrosis_vs_+SFI** as a single pre-specified secondary endpoint, its $p = 0.009$ against a family-wise threshold $\approx 0.05/12$ – $0.05/30$ is *suggestive, not confirmatory*.

Power analysis. In the synthetic cohort the subspace effect ($+0.0028$) needs $\sim 4,150$ cases for 80% power and the single-vector effect ($+0.0007$) $\sim 23,000$, clinically undetectable *there*. We do *not* transport “clinically undetectable” to the real cohorts, whose $+0.036$ – 0.051 estimates are compatible with a detectable effect we are simply underpowered to confirm.

4.2 The validity radius: why the single-vector estimator fails

The relative error of the first-order SFI scales as $\Theta(\rho)$ and crosses 10% at $\rho^* \approx 3$; real atria sit at median $\rho \approx 2422$. This proves the single-vector estimator is numerically invalid on real tissue and φ_2 is ill-conditioned, but (as boxed in Section 2.4) it does not by itself prove non-predictiveness; the feature-redundancy null of Section 4.1 is the separate, label-dependent reason.

4.3 The ρ scaling law

The gap collapses with system size by Weyl’s law [6]: for a d -dimensional domain the low Laplacian eigenvalues scale as $\lambda_k \sim (k/N)^{2/d}$, so ρ grows lawfully with resolution.

Table 2: Fitted gap-scaling exponents (gap vs. N at fixed mean degree, `results/rho_scaling.json`, five-point fits).

medium	fitted gap exponent	reference
real atrial surface (2-manifold)	$N^{-1.05}$	Weyl 2-manifold -1.0 (within 5%)
2-D random-geometric	$N^{-1.81}$	faster than naive $2/d$
3-D random-geometric	$N^{-0.84}$	faster than naive $2/d$ (non-monotone series)

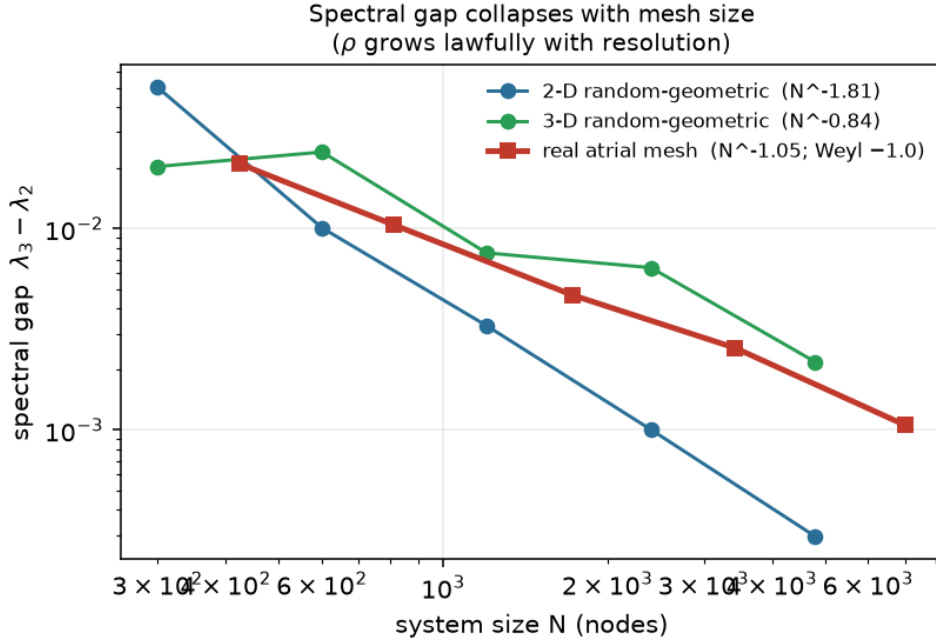


Figure 2: The gap collapses with mesh resolution, so the estimator degrades *more*, predictably, as the medium is meshed finer, the opposite of “more resolution helps.” Only the real atrial 2-manifold cleanly matches the Weyl exponent (-1.05 vs. -1.0); the random-geometric exponents are reported as “at least as fast as Weyl,” not law confirmations, and use the unweighted combinatorial Laplacian.

4.4 Localization transfers across three media (weakly)

The *strict* pre-registered localization endpoint is null for every field, including $|\nabla\varphi_2|$ (median argmax-geodesic error 0.472 is *not* below the spatial-null 5th percentile 0.406). What survives is

weaker: $|\nabla\varphi_2|$ is a *rank* localizer (the true origin sits in its top $\sim 18\%$) that beats graph-centrality baselines (Perron, weighted degree, deleted by measured correlation) and holds across media.

Table 3: Fiedler-gradient origin-rank localization across dynamical media (`results/gm4_*_metrics.json`). Lower rank is better; a perfect localizer would place the origin at rank 0.

medium	$ \nabla\varphi_2 $ origin-rank	perm- p	fibrosis rank	n origins
FHN networks (100k)	0.176	$< 10^{-5}$	~ 0.10	36,540
Kuramoto oscillators (500)	0.091	$< 10^{-4}$	0.147	243
cardiac monodomain	~ 0.28	$< 10^{-3}$	(co-localizes)	only ~ 20

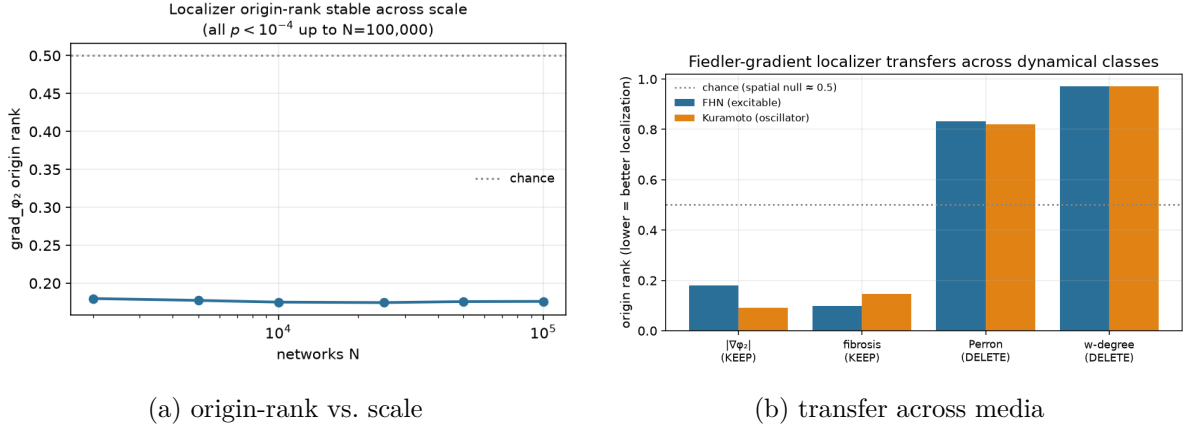


Figure 3: The weak-rank localization pattern replicates across excitable *and* oscillator dynamics. Three caveats apply: (i) the strict endpoint is null, so $|\nabla\varphi_2|$ is *not* a precise localizer; (ii) the substrate field co-localizes, and a controlled dissociation experiment shows why (in a diffusively-coupled medium the Fiedler bottleneck *is* the low-conduction lesion), so $|\nabla\varphi_2|$ matches, and does not beat, substrate imaging; (iii) the cardiac arm rests on only ~ 20 origins.

4.5 The falsification protocol: what caught the false positive

Each guard, removed, manufactures a specific false positive (`results/falsification_protocol.json`). Shape-family leakage inflates AUC by +0.117 (naive random-CV 0.756 vs. grouped 0.640 on a controlled replicated cohort); the effect-size gate catches the p -value trap (subspace-SFI $p \approx 2 \times 10^{-24}$ at $N = 55,000$, yet $\Delta\text{AUC} = +0.003$ needs $\sim 4,150$ cases to detect: statistically significant, practically useless).

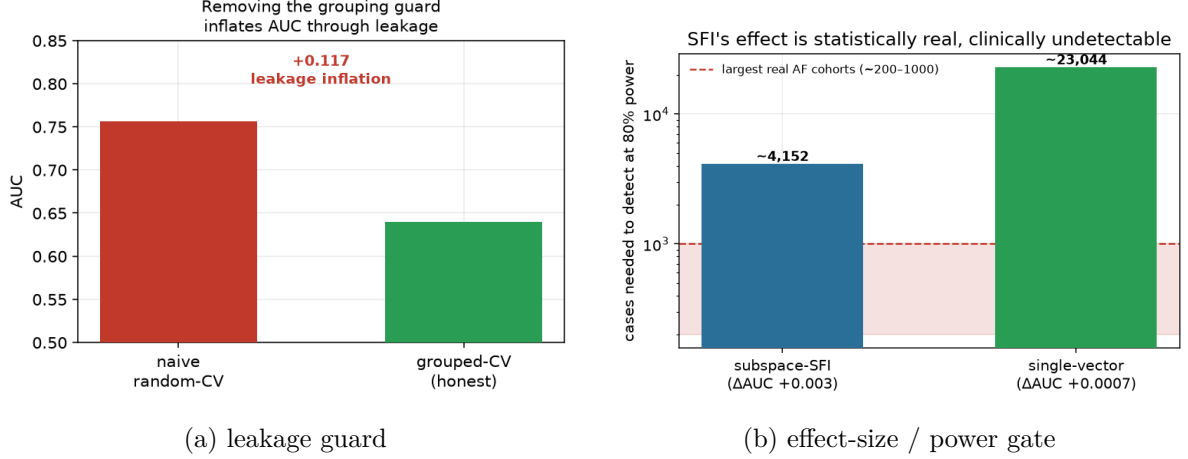


Figure 4: A naive analyst (random CV, $p < 0.05$, no effect-size gate, no spatial null) would have reported SFI as a working reentry biomarker. The pre-registered protocol correctly rejects it.

4.6 Uncertainty quantification

Morris elementary-effects screening (`results/e6_metrics.json`) finds the label most sensitive to diffusion d_0 ($\mu^* = 500.7$), then fibrosis density (147.8) and transverse conductivity (146.3). A Gaussian-process surrogate has leave-one-out $R^2 = 0.35$ (large residual anatomical/label variance beyond the screened knobs, a label-noise caveat, not a positive). The synthetic null is robust across uncoupling strengths $\Delta w \in \{0.18, 0.36, 0.54\}$ (grouped GBT $\Delta\text{AUC} = 0.033$ – 0.036 , all $p \geq 0.26$).

4.7 External inducibility-rate anchor (plausibility, not validation)

Clinically observed AF inducibility spans $\sim 5\%$ to $\sim 85\%$ depending on population, protocol aggressiveness, and the sustained-AF definition (Kumar et al. 2012 [14] report burst-sustained 14.8% vs. decremental-sustained 41.2% *in the same patients*). Our labeller’s Roney ~ 20 – 32% sits in the moderate-burst band (Marquardt et al. 2018 [15] report 30.6% in patients without prior AF), between the no-AF-control floor (5%, Oral et al. 2008 [16]) and the AF/aggressive ceiling (51%, Kawai et al. 2019 [17]), consistent with $\sim 42\%$ post-ablation (Liu et al. 2020 [18]). The UW $\sim 7\%$ is at/below the control floor for a cohort that *is* AF patients. We flag it as anomalous and attribute it to the substrate rather than to biology. Two substrate defects are documented in the Limitations, and repairing both lifts the rate to 17.1% — more than double, though still short of the 32.3% seen on fibre-carrying anatomy, so the anomaly is only partly accounted for and the in-silico inducer may also be under-aggressive. This is a base-rate *plausibility check*, not *clinical validation*.

4.8 Resolution convergence: the labeller does not converge in the tested window

Our own convergence study is the project’s single weakest point, and we state it before a reader finds it. On a completed six-tier study (Roney, $n = 24$ at each of 1500/3000/6000/12000/24000/48000 nodes; 144 simulations, `results/convergence_summary_roney.json`), the inducibility rate crashes from the coarse mesh (58.3% at 1500) to 16.7% at 3000, then is *non-monotone* across the fine tiers (25.0%, 20.8%, 25.0%) and rises again to 37.5% at the finest affordable tier ($\sim 50k$ nodes), a 17-percentage-point spread over the last three tiers, so it has *not* settled.

Table 4: Inducibility rate vs. mesh resolution (Roney, $n = 24$ per tier). The rate does not converge in the tested window; adjacent-tier per-subject verdict agreement climbs but the rate keeps drifting.

nodes	1,500	3,000	6,000	12,000	24,000	48,000
inducibility rate	58.3%	16.7%	25.0%	20.8%	25.0%	37.5%
adj.-tier agreement	—	50%	75%	71%	62%	79%

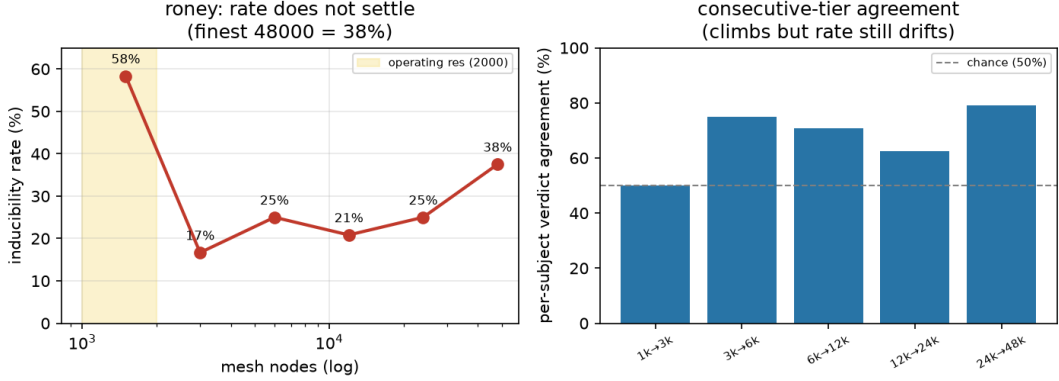


Figure 5: Left: the inducibility rate does not settle; the finest tier (48,000 nodes = 37.5%) sits above the mid tiers. Right: adjacent-tier per-subject verdict agreement climbs (50% → 79%) while the aggregate rate still drifts. Only 7/24 subjects are fully consistent across all six tiers.

The convergence finding limits the absolute numbers but not the comparative claims. The rate does *not* collapse to zero; the phenomenon is real; it is the *magnitude* that is resolution-sensitive up to $\sim 50k$ nodes. Because the main results ran at 2,000 nodes, every *absolute* number (inducibility rate, competitor AUC, localizer origins) is coarse-mesh-provisional pending the deferred openCARP validation. What this does *not* touch is what we lead with: (a) the *relative* SFI-vs-competitor comparison, since both arms score the *same* labels, so this bias cancels; and (b) the label-free estimator-validity argument (ρ is computed from the graph Laplacian, no labels at all).

5 Discussion

This work makes four contributions. (1) A pre-registered, leakage-controlled, effect-size-gated evaluation of a plausible spectral biomarker: a clean null at synthetic scale (10^5 networks) with its mechanism (feature redundancy: even the exact $\Delta\lambda_2$ SFI adds nothing), and an inconclusive, underpowered verdict on real anatomy. (2) A label-free validity criterion ρ for the single-vector estimator: it proves φ_2 is ill-conditioned on real tissue ($\rho \approx 2422 \gg \rho^* \approx 3$), a portable a priori check for any Fiedler-based marker, with the caveat that ρ bounds estimator accuracy, not predictive content. (3) A weak-but-real, purely connectivity-derived rank localizer that equals substrate localization and beats centrality baselines across three media. (4) A demonstration that standard methodology manufactures the exact false positives this protocol catches (leakage +0.117 AUC; the p -value trap).

The validity screen is domain-general. Spectral-graph fragility markers are used for seizure-focus localization, brain connectomics, and power-grid vulnerability. The same precondition applies: compute $\rho = \|\Delta L\|/(\lambda_3 - \lambda_2)$; if $\rho \gg 1$, the first-order spectral marker is outside its validity radius and its in-sample correlations cannot be trusted.

We are equally explicit about what we do not claim. The mathematics is classical; no new theorem. ρ does *not* prove SFI is non-predictive, only that the single-vector estimator is ill-conditioned; the null is separately due to feature redundancy. The real-cohort result is *not* a demonstrated null; it is underpowered. The localizer is *not* precise (strict endpoint null) and does *not* beat substrate imaging. The ρ scaling “law” is a clean match to Weyl only on the real atrial 2-manifold. Labels are a monodomain simulator, not clinical outcomes and not openCARP.

6 Limitations

1. The labeller over-calls inducibility at the operating mesh resolution and does not converge in the tested window (Section 4.8): the rate drops from 58% at 1500 nodes to 17% at 3000, is non-monotone across the fine tiers, and rises to 37.5% at $\sim 50k$ nodes; only 7/24 subjects are fully consistent across all six tiers. All *absolute* rates and AUCs are coarse-mesh-provisional; the *relative* comparison scores the same labels in both arms, so this bias cancels there.
2. Simulator, not clinical, labels: the pre-registered openCARP ground truth was substituted with monodomain Mitchell–Schaeffer, a logged deviation.
3. The UW cohort uses a PCA-surrogate universal-atrial-coordinate map, and its substrate carries two defects we identified only after the fact. The released meshes have a *degenerate* fibre field — the **VECTORS fiber** array is present but equals a constant (1,0,0) for every element of all 164 meshes, so the anisotropy the solver applies is a fixed coordinate bias rather than anatomy. Separately, we had been mapping element tag 164 to half-fibrotic tissue when it is in fact the set of caps over the pulmonary veins and the mitral valve. Across all 82 meshes it resolves into four to six large components, each a topological disc — five in 75 meshes, four in five of them and six in two, which is the usual variation in pulmonary-vein anatomy — and it borders fibrotic elements on only $\sim 0.08\%$ of its incident edges against a chance expectation of 14–20%. Removing it opens the surface, by a median of seven boundary loops. Left in, it lets activation cross those orifices instead of circling them. Repairing both raises inducibility from 6/82 to 14/82 ($7.3\% \rightarrow 17.1\%$), which does not reach paired significance (McNemar exact $p = 0.077$) and closes only 39% of the gap to the fibre-carrying Roney cohort (32.3%). A direct control settles the fibre half of that and overturns our first reading of it: destroying the fibre field on the Roney cohort, which ships a real one, leaves the inducibility rate *exactly unchanged* at 20/62 (McNemar $p = 1.0$), while still perturbing the dynamics — sustained-reentry duration moves on 50/62 subjects, by up to 661 ms. Fibre degeneracy therefore does not drive the inducibility *rate*, and our initial attribution of the UW anomaly to it was wrong. The anomaly stands unexplained. The categorical three-level fibrosis of these meshes, against Roney’s continuous 276–1071-level LGE field, is the leading remaining candidate. Either way the cohort’s predictive test is uninformative about SFI.
4. The inducibility label is stable in aggregate but noisy per subject, which attenuates every Δ AUC we report. Destroying the Roney fibre field holds the rate at 20/62 yet flips 10/62 individual verdicts; repairing the UW substrate retains only 2 of its 6 original positives. That compounds the mesh-convergence instability above, where just 7/24 subjects were consistent across six resolutions. Non-differential label noise biases incremental AUC toward zero, so the null we report is partly a statement about the labeller and not only about SFI.
5. ~ 30 DeLong comparisons were run; the one positive (**fibrosis_vs_+SFI**, $p = 0.009$) is a pre-specified secondary, suggestive not confirmatory.

6. ρ^* and the random-geometric gap exponents are empirical; only the real atrial 2-manifold cleanly matches the Weyl exponent.

7 Conclusion

A closed-form spectral fragility index does not add practically meaningful reentry-inducibility prediction beyond existing features at synthetic scale (a clean, feature-redundancy null), and on real anatomy the evidence is inconclusive rather than a demonstrated null. Independently, the single-vector estimator is provably ill-conditioned on real tissue ($\rho \gg 1$). The core result is not a discovery but a carefully-scoped boundary: where a Fiedler-based estimator is numerically valid, why real tissue lies far outside it, and how easily naive methodology would have mistaken redundant, inconclusive signal for a working biomarker. The absolute claims await the openCARP validation.

A Formal (proof-assistant) verification of the SFI algebraic core

The algebraic core of the SFI edge-sensitivity identity is re-proved inside the Coq 8.18 proof assistant and checked by its small trusted kernel, an assurance level above a computer-algebra system. The primary file (`formal/sfi_edge_identity_Q.v`) works over the exact rationals $\mathbb{Q}$, so `Print Assumptions` reports “*Closed under the global context*” for every theorem: zero axioms. A mathematically identical companion over the reals $\mathbb{R}$ (`formal/sfi_edge_identity.v`) inherits Coq’s two standard real-number-library axioms and is kept only to show the statements hold over $\mathbb{R}$.

Table 5: Kernel-checked theorems ($v = e_i - e_j$ the edge vector, $E_{ij} = \partial L / \partial w_{ij} = vv^\top$). T1–T5 are axiom-free over $\mathbb{Q}$.

name	statement	meaning
T1_rank_one_quadratic_form dot_edge (T2)	$\varphi^\top (vv^\top) \varphi = (v \cdot \varphi)^2$ $(e_i - e_j) \cdot \varphi = \varphi_i - \varphi_j$	quadratic form of a rank-one edge matrix the edge vector selects the potential difference
T3_sfi_edge_sensitivity	$\varphi^\top E_{ij} \varphi = (\varphi_i - \varphi_j)^2$	the SFI edge term
T4_sfi_term_nonneg	$0 \leq w \Rightarrow 0 \leq w(\varphi_i - \varphi_j)^2$	each SFI contribution is non-negative
T5_aggregate_nonneg	$\sum(\text{nonneg terms}) \geq 0$	SFI is a Dirichlet energy; the Laplacian form is PSD

Combined with the classical Hellmann–Feynman theorem $\partial \lambda_2 / \partial w_{ij} = \varphi_2^\top (\partial L / \partial w_{ij}) \varphi_2$ (Ghosh–Boyd 2006 [2], an analysis result we cite, not re-prove), T3 is exactly the closed form $\partial \lambda_2 / \partial w_{ij} = (\varphi_{2,i} - \varphi_{2,j})^2$. Lean/Mathlib was attempted; its toolchain binaries are distributed only as GitHub release assets, which the compute environment’s egress policy blocks with a hard 403, so Coq provides the equivalent kernel-checked guarantee. Reproduce with `coqc formal/sfi_edge_identity_Q.v`.

B Reproducibility

All stochastic steps are seeded; the exact configuration and commit hash accompany every reported number. Numeric verification: `python scripts/verify_sfi_identity.py` and `python scripts/verify_math_rigor.py` (sympy symbolic, mpmath 40-digit, exact-rational, interval-certified, and cross-solver checks). Convergence: `python scripts/convergence_analyze.py` roney. Figures are regenerated by `scripts/make_paper_figures.py`. Datasets: Roney LA

cohort (Zenodo 5801337) and UW/Boyle meshes (Dryad 10.5061/dryad.kkwh70sg0); both are public and de-identified.

References

- [1] M. Fiedler. Algebraic connectivity of graphs. *Czechoslovak Mathematical Journal*, 23(2):298–305, 1973.
- [2] A. Ghosh and S. Boyd. Growing well-connected graphs. In *Proc. 45th IEEE Conference on Decision and Control (CDC)*, 6605–6611, 2006.
- [3] L. M. Pecora and T. L. Carroll. Master stability functions for synchronized coupled systems. *Physical Review Letters*, 80(10):2109–2112, 1998.
- [4] M. Barahona and L. M. Pecora. Synchronization in small-world systems. *Physical Review Letters*, 89(5):054101, 2002.
- [5] W. Bomela, S. Wang, C.-A. Chou, and J.-S. Li. Real-time inference and detection of disruptive EEG networks for epileptic seizures. *Scientific Reports*, 10:8653, 2020.
- [6] H. Weyl. Das asymptotische Verteilungsgesetz der Eigenwerte linearer partieller Differentialgleichungen. *Mathematische Annalen*, 71(4):441–479, 1912.
- [7] C. Davis and W. M. Kahan. The rotation of eigenvectors by a perturbation. III. *SIAM Journal on Numerical Analysis*, 7(1):1–46, 1970.
- [8] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*, 44(3):837–845, 1988.
- [9] X. Sun and W. Xu. Fast implementation of DeLong’s algorithm for comparing the areas under correlated ROC curves. *IEEE Signal Processing Letters*, 21(11):1389–1393, 2014.
- [10] C. C. Mitchell and D. G. Schaeffer. A two-current model for the dynamics of cardiac membrane. *Bulletin of Mathematical Biology*, 65(5):767–793, 2003.
- [11] C. H. Roney et al. Predicting atrial fibrillation recurrence by combining population data and virtual cohorts of patient-specific left-atrial models. *Circulation: Arrhythmia and Electrophysiology*, 15(2):e010253, 2022. doi:10.1161/CIRCEP.121.010253. Data: Zenodo 5801337.
- [12] S. Bifulco, P. M. Boyle, et al. Predicting arrhythmia recurrence post-ablation in atrial fibrillation using explainable machine learning. *Communications Medicine*, 2025. doi:10.1038/s43856-025-01058-4. Data: Dryad 10.5061/dryad.kkwh70sg0.
- [13] I. M. Khurram et al. Magnetic resonance image intensity ratio, a normalized measure to enable interpatient comparability of left atrial fibrosis. *Heart Rhythm*, 11(1):85–92, 2014.
- [14] S. Kumar et al. Atrial fibrillation inducibility in the absence of structural heart disease or clinical atrial fibrillation: critical dependence on induction protocol, inducibility definition, and number of inductions. *Circulation: Arrhythmia and Electrophysiology*, 5(3):531–536, 2012. PMID 22528047.
- [15] J. Marquardt et al. Prognostic value of atrial fibrillation inducibility in patients without history of clinical atrial fibrillation. *Journal of Atrial Fibrillation*, 11(1):1837, 2018. PMID 30455834.

- [16] H. Oral et al. Inducibility of paroxysmal atrial fibrillation by isoproterenol and its relation to the mode of onset of atrial fibrillation. *Journal of Cardiovascular Electrophysiology*, 19(5):466–470, 2008. PMID 18266669.
- [17] S. Kawai et al. Predictive value of the induction test with atrial burst pacing with regard to long-term recurrence after ablation in persistent atrial fibrillation. *Journal of Arrhythmia*, 35(2):223–229, 2019. PMID 31007786.
- [18] X. Liu et al. Is atrial fibrillation noninducibility by burst pacing after catheter ablation associated with reduced clinical recurrence? A systematic review and meta-analysis. *Journal of the American Heart Association*, 9(14):e015260, 2020. PMID 32654581.